MIL-C-47092(MI)
24 May 1974
SUPERSEDING
MIS-15732
28 December 1965

MILITARY SPECIFICATION

COATING, POLYVINYL CHLORIDE, APPLICATION OF

This specification is approved for use by all activities of the Department of the Army.

1. SCOPE

1.1 Scope. This specification covers the application of a polyvinyl chloride coating to various surfaces such as steel, wood, concrete, and insulation materials. The coating is resistant to corrosion, abrasion, and most chemicals.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-47138	Coating, Polyvinyl Chloride

STANDARDS

Federal

FED-STD-141	Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling, and Testing

FSC-14GP

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:47Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Material. Polyvinyl chloride coating in accordance with MIL-C-47138.

3.2 Equipment.

3.2.1 Painting equipment. Suitable paint brushes and paint spray equipment.

3.2.2 Preparing equipment. Sand and vapor blast equipment.

3.3 Film thickness. Unless otherwise specified on the engineering drawing, three coats shall be applied and the total film thickness shall be between .003 and .0045 inches.

3.4 Porosity and adherence. The coating shall be firmly adhering, free of bubbles and not spongy or granular in appearance.

3.5 Temperature resistance. The fully cured coating shall exhibit no wrinkling, peeling, or other detrimental effects by continuous 250 degrees Fahrenheit (F) dry exposure or 155 degrees F wet immersion.

3.6 Procedure.

3.6.1 Surface preparation. All surfaces to be coated must be dry and free from dirt, grease and all loosely adhering material. Vapor degreasing or solvent wiping with a solvent such as methyl ethyl ketone, xylene, or toluene followed by wiping with a clean, ling-free cloth usually gives adequate cleaning. If necessary a light sand blast, liquid hone, or fine aluminum oxide cloth may be used as a step in surface preparation.

3.6.2 Application. The coating may be applied by brush or spray. Thin only as required to produce a uniform coating using the special thinner supplied by the manufacturer.

3.6.3 Coating thickness. The thickness of one brush or spray coat should be between .001 and .0015 inch. Several coats may be applied to bring the total thickness up to any desired level; however, each coat must be dry before the succeeding coat is applied. (See 3.3 and 4.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:47Z
Check the source to verify that this is the current version before use.

3.6.4 Drying time. The film dries tack-free in 10-15 minutes at room temperature; a minimum time of one hour should be allowed between coats at room temperature. Drying time can be reduced by increasing the temperature not to exceed 250 degrees Fahrenheit (F).

3.6.5 Curing. Maximum resistance to chemicals and moisture is not attained until all solvents have evaporated from the film. The required curing time depends on total film thickness, humidity and temperature. Raising the temperature will decrease the cure time. At room temperature the following will serve as a guide.

Film Thickness	Minimum Cure Time
.005 inch	3 days
.010 inch	5 days
.015 inch	8 days

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Film thickness. Film thickness shall be determined by measuring in accordance with Method 6183 of FED-STD-141.

4.3 Porosity and adherence. Visual inspection shall be used to establish conformity to 3.4.

4.4 Temperature resistance.

Source: https://assist.dla.mil -- Downloaded: 20261O-22T07:47Z
Check the source to verify that this is the current version before use.

4.4.1 Dry exposure. Dry exposure resistance shall be accomplished by subjecting the cured film to 250 plus or minus 5 degrees F for seven days. Visual inspection shall then determine conformity to 3.5.

4.4.2 Wet immersion. Resistance to wet immersion shall be determined by immersing the film in water held at 155 plus or minus 5 degrees F for seven days. Visual inspection shall determine conformity to 3.5.

5. PREPARATION FOR DELIVERY

This section is not applicable to this specification.

6. NOTES

6.1 Intended use. The application of polyvinyl chloride coating is intended for use in protective coating of various surfaces such as wood, steel, concrete, and insulation materials. The coating is resistant to corrosion, abrasion, and most chemicals.

6.2 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-15732, dated 28 December 1965.

Custodian:
Army-MI

Preparing Activity:
Army-MI

Project No. 14GP-A047

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:47Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:47Z
Check the source to verify that this is the current version before use.